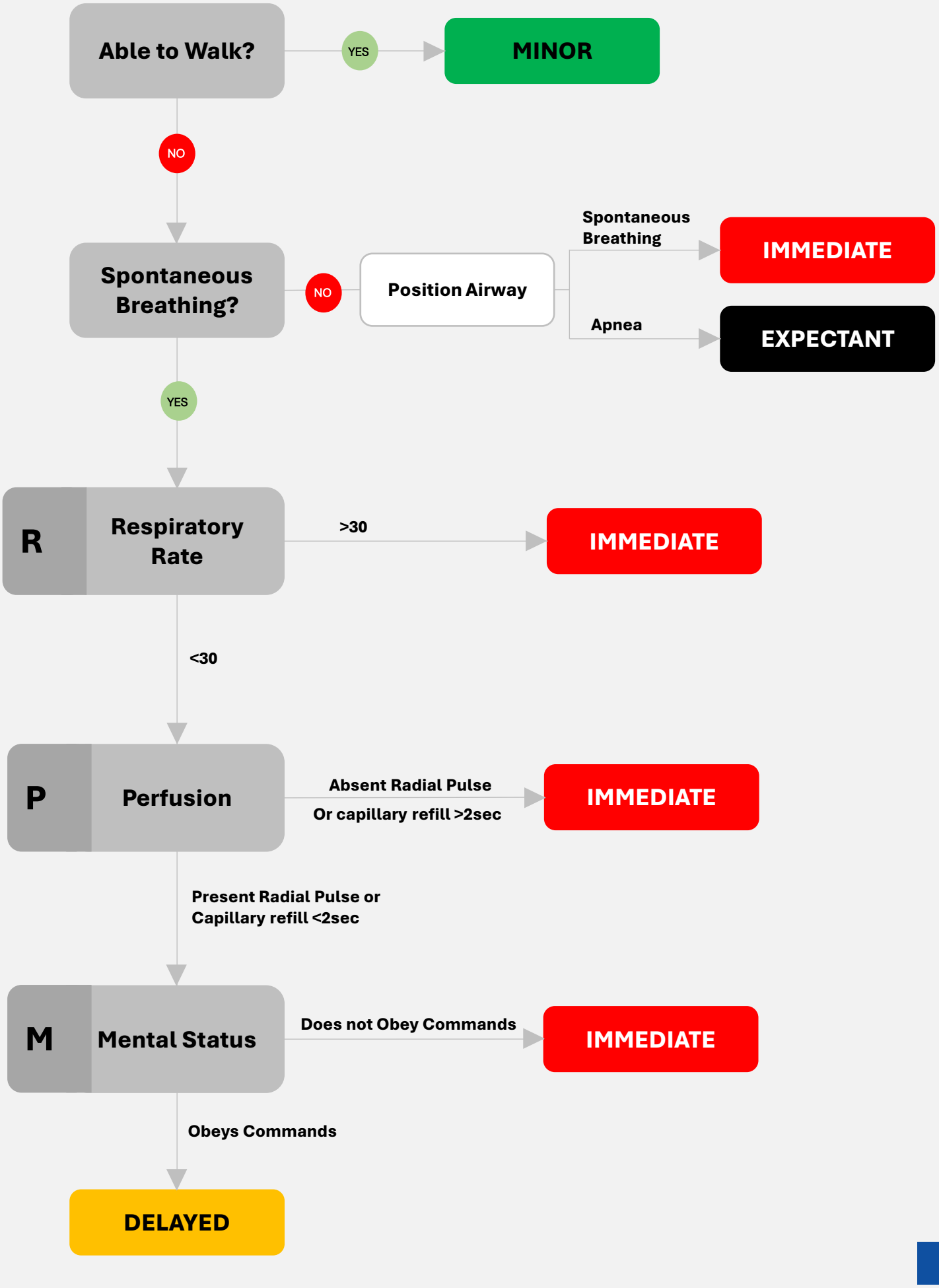
S.T.A.R.T Triage (MCI Triage)



Patient Care Goals

1. Rapidly categorize patients in mass casualty incidents (MCIs) to prioritize treatment and resource allocation based on injury severity and survivability.
2. Identify life-threatening conditions quickly to facilitate immediate interventions while maximizing the number of survivors.
3. Ensure efficient scene management and transport decisions based on patient priority.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients involved in mass casualty incidents (e.g., multi-vehicle accidents, explosions, natural disasters) requiring triage.
- All individuals at the scene, regardless of age, presenting with potential injuries or exposure.
- Symptomatic or asymptomatic patients needing rapid assessment for prioritization.

Exclusion Criteria

- Patients not part of a MCI.

Key Patient Assessment Points

1. **Sequential Evaluation** (START Algorithm)
    - Is the Patient Breathing Spontaneously?
        - Yes: Continue to assess Respiratory Rate
        - No: Position Airway, and reassess for Spontaneous Breathing:
            - No Spontaneous Breathing: Categorize as Expectant (Black)
            - Spontaneous Breathing: Continue to assess Respiratory Rate
    - Respiratory Rate Assessment:
        - Respiratory Rate > 30: Categorize as Immediate (Red)
        - Respiratory Rate < 30: Continue to Perfusion Assessment.
    - Perfusion Assessment:
        - Radial Pulse Absent or Capillary Refill >2seconds: Categorize as Immediate (Red)
        - Radial Pulse Present or Capillary Refill <2seconds: Continue to Mental Status Assessment.
    - Mental Status Assessment:
        - Doesn't obey commands: Categorize as Immediate (Red)
        - Obeys commands: Categorize as Delayed (Yellow)
2. **Category/Priority Assignment**
    - Immediate (Red): Requires urgent intervention.
    - Delayed (Yellow): Serious but not immediately life-threatening.
    - Minor (Green): Walking wounded; minimal injuries.
    - Expectant (Black): Unlikely to survive given resources.

Adult Patient Management & Interventions

2. Start with Global Command:
    - Start with global command:
        - "If you can hear me and are able to walk, come to my voice."
    - Categorize all walking/ambulatory patients as **Green Priority.**
    - Direct green priority patients to a designated area.
3. For non-ambulatory:
    - Apply basic interventions during triage: Airway positioning for apnea; control massive hemorrhage if identified.
    - **Follow Sequential Evaluation**
        - **Is the Patient Breathing Spontaneously?**
            - Yes: Continue to assess Respiratory Rate
            - No: Position Airway, and reassess for Spontaneous Breathing:
                - No Spontaneous Breathing: **Categorize as Expectant (Black)**

        - Spontaneous Breathing: Continue to assess Respiratory Rate
    - **Respiratory Rate Assessment:**
        - Respiratory Rate > 30: Categorize as **Immediate (Red)**
        - Respiratory Rate < 30: Continue to Perfusion Assessment.
    - **Perfusion Assessment:**
        - Radial Pulse Absent or Capillary Refill >2seconds: **Categorize as Immediate (Red)**
        - Radial Pulse Present or Capillary Refill <2seconds: Continue to Mental Status Assessment.
    - **Mental Status Assessment:**
        - Doesn't obey commands: **Categorize as Immediate (Red)**
        - Obeys commands: **Categorize as Delayed (Yellow)**
4. Integration with Incident Command
    - Report triage numbers to incident commander; adjust based on available resources.

Patient Safety Considerations

- Scene Safety:
    - Prioritize personal protective equipment; avoid entering unsafe areas during initial triage.

Clinical Insights & Additional Guidance

- Applying a triage tag to each patient is essential to ensure patients are treated in order of priority and to provide effective scene management.

Documentation

- Triage Tag Category Assigned should be documented by transferring team.

Key Performance Indicators

1. Complete triage assessment within 30 seconds (per patient)
2. Apply a triage tag to all patients triaged.

References

1. Emergency Department Triage - StatPearls. NCBI; 2023. Available at: https://www.ncbi.nlm.nih.gov/books/NBK557583/.
2. START Adult Triage Algorithm. REMM HHS; 2024. Available at: https://remm.hhs.gov/startadult.htm.
3. Accuracy of Triage Systems in Disasters and Mass Casualty Incidents. PMC; 2022. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC9078064/.
4. Comparison of START and SALT triage methodologies to reference standard definitions and to a field mass casualty simulation. American Journal of Disaster Medicine; 2017. Available at: https://wmpllc.org/ojs/index.php/ajdm/article/view/664.
5. Simple triage and rapid treatment protocol for emergency department overcrowding due to respiratory diseases. American Journal of Emergency Medicine; 2022. Available at: https://www.sciencedirect.com/science/article/abs/pii/S0735675721010081.
6. Mass casualty triage guidelines revised. Mayo Clinic; 2021. Available at: https://www.mayoclinic.org/medical-professionals/trauma/news/mass-casualty-triage-guidelines-revised/mac-20512735.
7. Assessment of three triage systems by medical undergraduate students in simulated mass casualty incidents. Frontiers in Disaster and Emergency Medicine; 2023. Available at: https://www.frontiersin.org/journals/disaster-and-emergency-medicine/articles/10.3389/femer.2023.1169851/full.
8. Disaster and Mass Casualty Triage. AMA Journal of Ethics; 2010. Available at: https://journalofethics.ama-assn.org/article/disaster-and-mass-casualty-triage/2010-06.
9. Triage Guidelines. CHEMM HHS; 2024. Available at: https://chemm.hhs.gov/triage.htm.
10. Triage: A Global Perspective. ScienceDirect; 2023. Available at: https://www.sciencedirect.com/science/article/abs/pii/S0099176723002143.

Scope of Practice Matrix